

BB AD Networking Written

How to Think. How to Approach. How to Practice.

From Theory to Real Exam Questions — Complete Mental Framework

The Problem	The Reality	The Solution
You read theory but cannot answer scenario questions	BB AD Written does NOT ask definitions — it asks you to APPLY concepts to situations	Learn the thinking pattern, not just the content

This guide breaks down every BB AD networking written question type into a thinking framework you can apply to any new question.

WHY THEORY IS NOT ENOUGH — Understanding the Real Problem

Before anything else, understand why students who read theory still cannot answer BB AD written questions.

What BB AD Written Actually Tests

Question Type	What It Really Asks	Theory Alone Helps?
Latency Calculation	Apply 4 delay formulas to a specific scenario with given numbers	No — you must practice calculation
Mail Transfer Steps	Trace the exact journey of an email using protocol knowledge	No — you must know the protocol flow
DNS Resolution	Describe what happens step-by-step when you type a URL	No — you must know the sequence
Server Access Types	Classify real-world server access using networking concepts	No — you must map concepts to scenarios
Subnetting Problem	Calculate subnet mask, hosts, network/broadcast addresses	No — you must practice problems
Protocol Identification	Name Application + Transport protocols for a given scenario	Partial — need to apply, not just list
Parity + ASCII	Write ASCII bit strings and apply parity error detection	No — you must know ASCII table and method

The Real Difference: Definition vs Application

Theory tells you: "DNS resolves domain names to IP addresses."

BB AD asks: "Afsana types www.example.bb.org in her browser. Her DNS is unknown. Write the exact steps of DNS name resolution."

Theory tells you: "There are 4 types of delay in networking."

BB AD asks: "A 3KB email is sent over a 1 Gbps network. Distance = 300km, speed = 2×10^8 m/s, RTT = 50ms, Queue delay = 5ms. Calculate total latency."

HOW TO THINK: BB AD tests whether you can USE the theory, not whether you can REPEAT it. The mental shift needed: After reading any concept, immediately ask yourself — "Given a specific scenario with specific numbers/names, how would I APPLY this?"

01 THE 6 TYPES OF BB AD NETWORKING WRITTEN QUESTIONS

Every BB AD networking written question falls into one of these 6 types. Once you identify the type, you know exactly what to do.

Type #	Question Pattern	What You Need to DO	Years It Appeared
Type 1	Calculate [latency/delay/throughput/source rate] given [numbers]	Identify which formula applies. Plug in numbers. Show each step.	2025, 2023, 2022, 2021
Type 2	Write the steps of [protocol/process] from A to B	Trace the protocol journey step-by-step in sequence.	2025, 2017, 2021
Type 3	What kind of access does [this server] require?	Classify using: public/private, static/dynamic, inbound/outbound.	2025, 2019
Type 4	Given [IP/subnet], find [network address/hosts/subnets/broadcast]	Apply subnetting formulas. Show all working.	2017, 2022, 2021, 2020
Type 5	Name the [Application/Transport] layer protocols for [scenario]	Map scenario to OSI layers and identify protocols.	2025, 2019
Type 6	Explain [concept] with a diagram or short note	Structured explanation: definition + how it works + example.	2019, 2017, 2016, 2011

KEY POINT: Most BB AD networking written questions are Type 1 (Calculation) or Type 2 (Steps/Protocol flow). These 2 types alone account for 70%+ of networking written marks. Master them first.

02 TYPE 1 — CALCULATION QUESTIONS (Most Common)

BB AD 2025 Q3 was a latency calculation. BB AME 2023 Q2 and Q3 were source rate and packetization delay. This type appears EVERY year. You must master it.

2A — The Latency Calculation Framework

HOW TO THINK: When you see numbers like: distance, bandwidth, message size, RTT, queue delay — it is a LATENCY question. Your job is to:

- (1) Identify which delays are given/calculable,
- (2) Apply the right formula for each,
- (3) Add them up. Never start writing until you have listed all given values.

Step 1: Read the question and extract ALL given values

Write these down immediately — do not hold them in your head: Message size: _____ (bytes or bits?) Bandwidth: _____ (bps, Kbps, Mbps, Gbps?) Distance: _____ (km or m?) Signal speed: _____ (m/s) RTT: _____ (ms or s?) Queuing delay: _____ (given directly or not?) Processing delay: _____ (given or derive from RTT?)

Step 2: Convert ALL units BEFORE calculating

Message: bytes → bits (multiply by 8) Bandwidth: Gbps = 10^9 , Mbps = 10^6 , Kbps = 10^3 Distance: km → m (multiply by 1000) 1 KB (network) = 1000 bytes = 8000 bits [NOT 1024!] All final delays in same unit (ms or seconds)

Step 3: Calculate each delay separately, LABEL each one

$$\begin{aligned} \text{Transmission Delay} &= \text{Message size (bits)} \div \text{Bandwidth (bps)} \\ \text{Propagation Delay} &= \text{Distance (m)} \div \text{Signal speed (m/s)} \\ \text{Processing Delay} &= (\text{RTT} \div 2) - \text{Propagation Delay} \\ \text{Queuing Delay} &= \text{Given directly in the problem} \end{aligned}$$

Step 4: Write Total Latency = sum, show the addition

$$\text{Total Latency} = \text{Prop} + \text{Trans} + \text{Queue} + \text{Processing}$$

QUESTION: BB AD 2025 Written Q3 (Exact): A 3-kbyte email is sent over a network with 1 Gbps bandwidth. Distance = 300 km, signal speed = 2×10^8 m/s. RTT = 50 ms, Queuing delay = 5 ms. Find the total latency.

HOW TO THINK: Read question → extract values → identify what is given directly vs what needs calculation → Processing delay is NOT given, so derive from RTT. Queue delay IS given. Transmission and Propagation need calculation.

MODEL ANSWER — BB AD 2025 Latency — Full Working

STEP 1 — Extract given values:

- Message size = 3 KB = $3 \times 1000 \times 8 = 24,000$ bits
- Bandwidth = 1 Gbps = 1×10^9 bps
- Distance = 300 km = 300,000 m
- Signal speed = 2×10^8 m/s
- RTT = 50 ms
- Queuing delay = 5 ms (given directly)

STEP 2 — Calculate each delay:

- Transmission Delay = $24,000 \div (1 \times 10^9) = 0.000024 \text{ s} = 0.024 \text{ ms}$
- Propagation Delay = $300,000 \div (2 \times 10^8) = 0.0015 \text{ s} = 1.5 \text{ ms}$
- Processing Delay = $(\text{RTT} \div 2) - \text{Propagation Delay} = 25 - 1.5 = 23.5 \text{ ms}$
- Queuing Delay = 5 ms

STEP 3 — Total Latency:

- = Transmission + Propagation + Processing + Queuing
- = $0.024 + 1.5 + 23.5 + 5 = 30.024 \text{ ms}$

Answer: Total Latency $\approx 30 \text{ ms}$

2B — Source Rate Calculation (BB AME 2023)

HOW TO THINK: When you see: pixels, intensity levels, frames per second → it is a SOURCE RATE question. The formula comes from: how many bits does one frame need? Then multiply by frames per second.

QUESTION: BB AME 2023: A digitized TV picture has a 480×500 pixel matrix. Each pixel has 32 intensity values. 30 pictures/sec transmitted. Find source rate R.

MODEL ANSWER — BB AME 2023 Source Rate

- **STEP 1 — Bits per pixel:** Intensity levels = 32 → bits = $\log_2(32) = 5$ bits per pixel
- **STEP 2 — Bits per frame:** = $480 \times 500 \text{ pixels} \times 5 \text{ bits} = 1,200,000$ bits per frame
- **STEP 3 — Source Rate:** $R = 1,200,000 \times 30 \text{ frames/sec} = 36,000,000 \text{ bps} = 36 \text{ Mbps}$

2C — Packetization Delay (BB AME 2023)

HOW TO THINK: When you see: packet size (bytes), header size (bytes), source encoding rate (kbps) → it is a PACKETIZATION DELAY question. The key insight: delay = time to fill the payload portion of one packet.

QUESTION: BB AME 2023: Voice source encoded at 128 kbps. Packet has P=1500 bytes total with 5-byte header. Find packetization delay.

MODEL ANSWER — BB AME 2023 Packetization Delay

- **STEP 1 — Payload size:** Payload = (1500 – 5) bytes = 1495 bytes = $1495 \times 8 = 11,960$ bits
- **STEP 2 — Packetization Delay:** = Payload (bits) ÷ Source Rate (bps) = $11,960 \div 128,000 = 0.0934 \text{ s} \approx 93.4 \text{ ms}$

PRACTICE: PRACTICE PLAN for Type 1:

Solve 2 latency problems per day for 5 days. Variation drills:

- (a) Only propagation and transmission given — calculate those 2 only.
- (b) RTT given — derive processing.
- (c) Source rate given — find packetization.
- (d) BDP question. Always write every step with labels.

03 TYPE 2 — PROTOCOL STEPS (Second Most Common)

BB AD 2025 Q7 asked mail transfer steps. Type 2 questions test whether you can trace a protocol JOURNEY — not define what it is. The answer must be in numbered steps, in exact sequence.

3A — The Protocol Steps Thinking Framework

HOW TO THINK: Any protocol steps question has: a SENDER, a RECEIVER, and INTERMEDIATE NODES. Your job is to trace the journey from sender to receiver, naming every stop, every protocol used, every action taken. Think like a packet traveling through the network.

General Template for Protocol Steps Answers:

Step 1: Identify WHO sends, WHO receives, and WHAT is being sent
Step 2: Identify the protocols involved at each stage (Application/Transport/Network)
Step 3: Trace the journey from source to destination naming every node
Step 4: Include what happens AT each node (lookup, forward, deliver, etc.)
Step 5: Include the reverse journey if it is request-response (like DNS)

3B — Mail Transfer Steps (BB AD 2025 Q7)

QUESTION: BB AD 2025 Written Q7: Afsana (www.example.bb.org) and Sinthia (www.example.org.uk) exchange email (unknown DNS). (a) Name Application and Transport Layer protocols. (b) Write down the steps of mail transfer from Afsana to Sinthia.

HOW TO THINK: This is NOT asking "what is SMTP?" It is asking you to TRACE the email journey. Start from Afsana composing the email and end with Sinthia reading it. Every step must name the protocol and what it does.

MODEL ANSWER — BB AD 2025 Q7 — Full Answer**(a) Protocols**

- **Application Layer:**
 - SMTP (Simple Mail Transfer Protocol) — for sending
 - DNS (Domain Name System) — for MX record lookup
 - POP3 or IMAP — for Sinthia to retrieve email
- **Transport Layer:**
 - TCP (port 25 for SMTP, port 110 for POP3, port 143 for IMAP)

(b) Mail Transfer Steps — Afsana to Sinthia

1. **Afsana composes email:** Afsana uses a Mail User Agent (MUA) (e.g., Outlook or Gmail web client). She types "To: sinthia@example.org.uk" and writes the message body.
2. **MUA submits to Afsana's Mail Server (MTA):** MUA sends the email to Afsana's outgoing Mail Transfer Agent (MTA) using SMTP protocol over TCP port 587 (or 25).
3. **Afsana's MTA does DNS MX record lookup:** Afsana's MTA queries DNS: "What is the mail server (MX record) for example.org.uk?" DNS responds with: mail.example.org.uk = 203.x.x.x (Sinthia's mail server IP).
 - *Note: Since DNS is unknown, the MTA uses iterative DNS resolution: Root DNS → .uk TLD DNS → example.org.uk Name Server → MX record returned.*
4. **Afsana's MTA connects to Sinthia's MTA:** Afsana's MTA establishes TCP connection to Sinthia's MTA on port 25. Transfers the email using SMTP commands: HELO, MAIL FROM, RCPT TO, DATA, QUIT.
5. **Sinthia's MTA stores email in mailbox:** Sinthia's MTA (receiving MDA — Mail Delivery Agent) accepts the message and stores it in Sinthia's mailbox on the mail server.
6. **Sinthia retrieves her email:** Sinthia uses her MUA (email client) to connect to her mail server.
 - Uses POP3 (port 110) to download and delete from server, OR
 - Uses IMAP (port 143) to read while keeping on server (sync across devices).
7. **Sinthia reads the email in her MUA.**

3C — DNS Resolution Steps (BB Written 2025 + 2026)

QUESTION: Other Bank 2025/2026: Why does DNS use UDP instead of TCP? Write the exact DNS name resolution steps when a user types www.companybd.com into a browser.

HOW TO THINK: DNS steps question = trace the query from user's browser to the IP address. There are 4 servers involved: Local cache, Recursive Resolver, Root DNS, TLD DNS, Authoritative DNS. Know the path.

MODEL ANSWER — DNS Resolution — Full Step-by-Step**Why UDP?**

- DNS queries are small (under 512 bytes typically).
- UDP is faster — no connection setup overhead.
- If response is lost, DNS simply retries.
- DNS switches to TCP only for large responses (zone transfers > 512 bytes).

DNS Resolution Steps for www.companybd.com:

- 1. Browser checks local DNS cache:** Browser looks for www.companybd.com in its own cache. If found and TTL not expired → use cached IP → done.
- 2. OS checks its DNS cache:** If not in browser cache, OS checks its resolver cache. If found → return IP to browser.
- 3. Query sent to Recursive Resolver (ISP DNS):** OS sends UDP query to configured DNS server (e.g., 8.8.8.8 or ISP DNS). Recursive resolver checks its own cache first.
- 4. Recursive Resolver queries Root DNS Server:** Resolver asks Root DNS: "Who handles .com domains?" Root DNS responds: "Ask the .com TLD server at 192.5.6.30"
- 5. Recursive Resolver queries .com TLD Server:** Resolver asks .com TLD: "Who is authoritative for companybd.com?" TLD responds: "Ask ns1.companybd.com at 203.x.x.x"
- 6. Recursive Resolver queries Authoritative Name Server:** Resolver asks companybd.com name server: "What is the IP of www.companybd.com?" Authoritative server responds with the A record: 103.x.x.x
- 7. IP returned to browser:** Recursive resolver caches the result (per TTL) and returns IP to browser. Browser connects to 103.x.x.x via HTTP/HTTPS. Browser caches the result for future requests.

PRACTICE: PRACTICE for Type 2: Write the steps of (a) SMTP mail transfer (b) DNS resolution (c) DHCP IP assignment (DORA) (d) HTTP request-response — each from memory without notes. Then compare with the model answers. The goal: write 6-7 steps in correct sequence within 5 minutes.

04 TYPE 3 — SERVER ACCESS CLASSIFICATION (BB AD 2025 Q8)

QUESTION: BB AD 2025 Written Q8: Mail Server, DNS Server, Public Website — what kind of access do they require?

HOW TO THINK: This question looks simple but confuses many students. The key framework: think about WHO accesses it, from WHERE, and HOW. Ask 3 questions: (1) Internal-only or Internet-facing? (2) Static IP or Dynamic? (3) Inbound from outside or Outbound only?

Framework for Server Access Classification

Question to Ask	Options	Determines
Who accesses this server?	Internal users only / External (internet) users / Both	Network placement: LAN / DMZ / Public cloud
Does it need a static IP?	Yes (always same IP) / No (dynamic OK)	IP assignment type: static or DHCP
Direction of traffic?	Inbound (internet → server) / Outbound (server → internet) / Both	Firewall rules needed
Does it need public reachability?	Yes (must be reachable from internet) / No (private only)	Public IP or private IP with NAT

MODEL ANSWER — BB AD 2025 Q8 — Server Access Types

Mail Server (e.g., Afsana's mail.example.bb.org)

- **Access Type:** Both internal AND external (internet-facing)
- **Location:** Typically in DMZ (Demilitarized Zone)
- **IP:** Static public IP — must be reachable from internet for email delivery
- **Inbound:** Port 25 (SMTP) from other mail servers
- **Outbound:** Port 25 to send emails to other mail servers
- **Firewall:** Allow inbound TCP port 25 from internet to mail server

DNS Server

- **Types:** Internal DNS (for LAN) and External/Authoritative DNS (for internet)
- **External DNS:** Must be publicly accessible with a static public IP
- **Location:** DMZ or hosted by DNS provider
- **Access:** Responds to UDP/TCP port 53 queries from anyone on internet
- **Inbound:** Port 53 from anywhere (UDP for queries, TCP for zone transfers)

Public Website (Web Server)

- **Access Type:** External (anyone on internet can access)
- **Location:** DMZ or cloud hosting
- **IP:** Static public IP (or CDN with domain pointing to IP)
- **Inbound:** TCP port 80 (HTTP) and TCP port 443 (HTTPS) from internet
- **SSL/TLS:** Certificate required for HTTPS
- **Firewall:** Allow inbound 80 and 443, block all other ports

KEY CONCEPT — DMZ

- Servers that need to be internet-accessible (web, email, DNS) go in DMZ.
- DMZ is protected from internet by external firewall AND from internal LAN by internal firewall.
- This way, even if web server is compromised, internal LAN remains safe.

PRACTICE: PRACTICE for Type 3: For each of these servers, classify access type: FTP Server, Database Server, Internal File Server, VPN Server, Active Directory Server. Use the 4-question framework for each.

05 TYPE 4 — SUBNETTING PROBLEMS

Subnetting is pure calculation. Once you know the 3 formulas, any subnetting question is solvable. The challenge is not the formula — it is reading the question carefully and identifying exactly what is being asked.

Subnetting Question Reading Framework

HOW TO THINK: Before calculating, identify WHAT is given and WHAT is asked: (1) Is the prefix given or do I need to find it? (2) Am I finding total addresses, usable hosts, or number of subnets? (3) Do I need network address and broadcast? (4) Is it VLSM (different size subnets)?

If the question gives...	Then you need to find...	Formula to use
IP/prefix like 172.20.0.0/27	Total IPs, usable hosts, subnets	$h=32-\text{prefix}$, 2^h , 2^{h-2} , subnets= 2^{borrowed}
IP and "needs X subnets"	New prefix, subnet mask, hosts per subnet	Borrow n bits where $2^n \geq X$
IP and "needs X hosts per subnet"	Prefix, subnet mask, total subnets	Find h where $2^{h-2} \geq X$, prefix= $32-h$
IP/prefix and specific host number	Which subnet it falls in, network+broadcast	Block size method

QUESTION: BB Written 2017: How many subnets and hosts per subnet from 172.20.0.0/27?

MODEL ANSWER — BB 2017 Subnetting Class B address (default /16). New prefix = /27. Borrowed bits = $27 - 16 = 11$
 Subnets = $2^{11} = 2048$
 Host bits = $32 - 27 = 5$
 Usable hosts = $2^5 - 2 = 30$
 Subnet mask = 255.255.255.224

QUESTION: Other Bank 2026: Given IP 192.168.1.0, divide into 4 equal subnets. (A) New subnet mask (CIDR). (B) First usable host of each subnet.

MODEL ANSWER — Other Bank 2026 Subnetting — 4 Equal Subnets

- **Requirements:** 4 subnets needed → borrow 2 bits ($2^2 = 4$)
- **New Prefix:** $/24 + 2 = /26$
- **New Mask:** 255.255.255.192
- **Block Size:** $2^6 = 64$

Subnet	Network Address	First Usable Host	Broadcast Address
1	192.168.1.0	192.168.1.1	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.255

PRACTICE: PRACTICE for Type 4: Solve 3 subnetting problems per day. Mix these types:

- Given prefix, find subnets+hosts.
- Given host requirement, find prefix.
- Given IP, find which subnet it belongs to.
- Divide one network into N equal subnets. Always write all working — never just the answer.

06 TYPE 5 — PROTOCOL IDENTIFICATION FOR A SCENARIO

QUESTION: BB AD 2025: Afsana and Sinthia at www.example.bb.org and www.example.org.uk. Name the Application Layer and Transport Layer protocols.

HOW TO THINK: Protocol identification requires mapping a scenario to OSI layers. The mental process: What ACTION is happening? (email → SMTP/POP3/IMAP, web → HTTP/HTTPS, file → FTP, remote → SSH/Telnet). What layer does it run on? What does TCP/UDP provide underneath?

Scenario → Protocol Mapping Reference

Scenario/Action	Application Layer Protocol	Transport Layer Protocol
Sending email	SMTP (port 25/587)	TCP
Receiving email (download)	POP3 (port 110)	TCP
Reading email (keep on server)	IMAP (port 143)	TCP
Web browsing (unencrypted)	HTTP (port 80)	TCP
Secure web browsing	HTTPS (port 443) with TLS/SSL	TCP
Uploading/downloading files	FTP (port 20/21)	TCP
Domain name lookup	DNS (port 53)	UDP (small), TCP (large)
Auto IP assignment	DHCP (port 67/68)	UDP
Secure remote login	SSH (port 22)	TCP
Remote login (insecure)	Telnet (port 23)	TCP
Network device monitoring	SNMP (port 161/162)	UDP
Voice over internet	VoIP / SIP / RTP	UDP (for voice)
Video streaming	RTSP / RTP	UDP
Finding MAC from IP	ARP	No transport (Layer 2-3)

MODEL ANSWER — BB AD 2025 — Protocol Identification Answer

Scenario: Afsana sends email to Sinthia. DNS lookup needed (unknown DNS).

Application Layer Protocols:

- SMTP — Simple Mail Transfer Protocol (port 25) — to SEND the email
- DNS — Domain Name System (port 53) — to find the MX record of recipient domain
- POP3 — Post Office Protocol 3 (port 110) — for Sinthia to RECEIVE the email
- OR IMAP — Internet Message Access Protocol (port 143) — if Sinthia uses multiple devices

Transport Layer Protocols:

- TCP — for SMTP, POP3, IMAP (reliable delivery needed for email)
- UDP — for DNS queries (fast, small packets; retried if lost)

Note: SMTP between mail servers uses port 25. From client to server (submission) uses port 587.

07 TYPE 6 — STRUCTURED SHORT EXPLANATIONS

These questions ask you to explain a concept clearly with structure. They appear as: "Write short notes on...", "Explain with diagram...", "Differentiate between..."

Template for Structured Explanations

For "What is X?" questions: Para 1: Definition (1-2 sentences) — what it IS Para 2: How it works — mechanism/steps Para 3: Example — concrete real-world case Para 4: (If asked) advantages or purpose
For "Differentiate X and Y" questions: Use a TABLE with 3-4 rows: Definition, How it works, Example, Use case Never write paragraphs for comparison — always use a table

QUESTION: BB Written 2019: What is a Firewall? Explain its function. Draw a LAN diagram showing firewall placement.

MODEL ANSWER — BB 2019 Firewall

Definition:

A firewall is a network security device (hardware or software) that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks (Internet).

How it Works:

- **Rule Matching:**
 - 1. If packet matches ALLOW rule → packet is forwarded.
 - 2. If packet matches DENY rule → packet is dropped/rejected.
 - 3. If no rule matches → default action (usually deny).
- **Types:**
 - Packet-filter: Checks IP/port only.
 - Stateful: Tracks connection state.
 - NGFW: Deep packet inspection, application awareness.

Diagram (Placement):

- Internet → [External Firewall] → [DMZ] (Public servers: Web server, Mail server, DNS server) → [Internal Firewall] → [Internal LAN] (Private: File servers, Databases, Employee workstations).

Purpose of DMZ:

- Servers in the DMZ are accessible from the Internet but isolated from the internal LAN.
- Even if the web server is compromised, the internal LAN remains protected by the internal firewall.

QUESTION: BB Written 2017: Write short notes on DHCP and SMTP.

MODEL ANSWER — Short Notes Template — DHCP and SMTP

DHCP (Dynamic Host Configuration Protocol): Definition: A network protocol that automatically assigns IP addresses and other network configuration (subnet mask, gateway, DNS) to devices when they join a network. Port: 67 (server), 68 (client).

Transport: UDP. How it works (DORA process): D = Discover: Client broadcasts "I need an IP" (no IP yet) O = Offer: DHCP server offers an available IP with lease time R = Request: Client accepts the offered IP A = Acknowledge: Server confirms and records the assignment Benefit: Eliminates manual IP configuration. Prevents IP conflicts. **SMTP (Simple Mail Transfer Protocol):** Definition: A protocol used to SEND email messages between mail servers and from mail clients to mail servers. Port: 25 (server-to-server), 587 (client-to-server). Transport: TCP. How it works: 1. Sender's email client connects to outgoing mail server using SMTP 2. Sender's MTA performs DNS MX lookup for recipient domain 3. Sender's MTA connects to recipient's MTA and transfers email via SMTP 4. Recipient retrieves email using POP3 or IMAP (not SMTP) Key point: SMTP is for SENDING only. POP3/IMAP are used for RECEIVING.

08 HOW TO WRITE YOUR ANSWERS IN THE EXAM HALL

Knowing the answer is not enough. How you WRITE it determines how many marks you get. BB AD written uses partial marking — structured answers get more marks even if you miss some steps.

The 3-Minute Planning Rule

HOW TO THINK: Before writing ANY answer, spend 2-3 minutes planning on rough paper: (1) What TYPE is this question? (2) What are all the things I need to include? (3) In what ORDER? Only then start writing. This prevents the most common mistake: starting to write and then realizing you forgot something important halfway through.

Formatting Rules for Maximum Marks

Question Type	Best Format	What Examiners Look For
Calculation (Type 1)	Show: Given values → Each formula → Each calculation → Final total	Every step labeled. Correct formula. Correct unit conversions. Final answer with unit.
Protocol Steps (Type 2)	Numbered list: Step 1, Step 2... each with protocol name	Correct sequence. Protocol names. What happens at each step. Minimum 5-6 steps.
Server Classification (Type 3)	Bold server name, then bulleted properties	Access type, IP type, ports, firewall rules, location (DMZ/LAN).
Subnetting (Type 4)	Show formula → calculation → final answer	All formulas visible. Correct h value. Correct subtraction of 2. Block size shown.
Protocol ID (Type 5)	Table: Layer Protocol Port Reason	Both Application and Transport layers covered. Port numbers included.
Explanation (Type 6)	Definition → How it works → Example → Diagram	Structured paragraphs or table. Diagram adds 2-3 marks. Concrete example.

Do NOT Do These in Your Answer

- **Never start with "I will explain..."** — Start directly with the content
- **Never write a paragraph for calculations** — Write each step on a new line with the formula
- **Never write steps as one long paragraph** — Number every step
- **Never leave the unit out** — Always write ms, bps, m/s, bits with every answer
- **Never write "etc."** — Be specific, name the actual protocols/devices
- **Never skip showing the formula** — Write the formula THEN substitute numbers

Partial Marking Strategy — Never Leave Blank

KEY POINT: BB AD Written has PARTIAL marking. If a 10-mark question has 5 steps and you know 3, write those 3 — you get 6 marks instead of 0. If you cannot complete a calculation, write the formula and the setup — that alone earns marks. If you cannot draw a diagram, describe it in words. NEVER leave any question blank.

09 14-DAY PRACTICE SCHEDULE — From Theory to Exam Ready

This schedule takes you from knowing the theory to being able to answer any BB AD networking written question under exam conditions.

Day	Focus	What to Do	Time
Day 1	Latency Setup	Read Section 2. Write all 4 delay formulas from memory 3 times.	1 hr
Day 2	Latency Practice	Solve: BB 2025 Q3 without looking at answer. Then check. Note mistakes.	1 hr
Day 3	More Latency + Source Rate	Solve 3 new latency problems. Solve BB AME 2023 source rate + packetization.	1.5 hr
Day 4	Mail Transfer Steps	Write SMTP mail transfer steps from memory (7 steps). Compare. Repeat.	1 hr
Day 5	DNS Steps	Write DNS resolution steps from memory (7 steps). Write WHY DNS uses UDP.	1 hr
Day 6	DHCP + Protocols	Write DORA steps. Write protocol table from Section 6 from memory.	1 hr
Day 7	Mixed Practice Test	Without notes: (a) Latency calculation (new numbers) (b) Mail steps (c) Protocol ID	1.5 hr
Day 8	Subnetting Setup	Review subnetting formulas. Solve 5 problems (different types each).	1.5 hr
Day 9	Subnetting Practice	Solve: BB 2017, Other Bank 2026 4-subnet, Other Bank ANE JBL 2022.	1.5 hr
Day 10	Server Access + Firewall	Write BB 2025 Q8 answer structure. Draw DMZ diagram from memory.	1 hr
Day 11	Structured Explanations	Write DHCP + SMTP short notes. Write Firewall explanation. No notes.	1 hr
Day 12	Full Mixed Practice	Solve: 1 latency + 1 protocol steps + 1 subnetting — timed 45 minutes.	1.5 hr
Day 13	All BB AD Questions	Solve ALL BB AD networking written questions from Section 15 of guide.	2 hr
Day 14	Exam Simulation	50 min timed: Write answers to 3 networking questions (no notes, no help).	1 hr

APPROACH: The most important days are Day 7, Day 12, and Day 14 — the timed practice sessions. Do NOT skip these. Writing under time pressure is a different skill from understanding the concept. You need both.

How to Review Your Practice Answers

- **Step 1:** Check every formula used — is it correct?
- **Step 2:** Check every unit conversion — bytes to bits? km to m?
- **Step 3:** Check the sequence of steps — is anything missing or in wrong order?
- **Step 4:** Check protocol names — exact name and port number?
- **Step 5:** Time yourself — can you complete this in exam time?
- **Step 6:** Identify 1 thing to improve and practice that specifically

Quick Reference — The 6 Types Summary

Type	Trigger Words	First Thing to Do
Type 1 Calculation	calculate, find, determine, how long, what is the latency/delay/rate	Write: Given: and extract ALL numbers first
Type 2 Protocol Steps	write the steps, describe the process, how does X transfer	Write: Step 1, Step 2... trace the full journey
Type 3 Server Access	what kind of access, what does X server require	Classify: internal/external? Static IP? Which ports?
Type 4 Subnetting	subnets, hosts per subnet, network address, broadcast, divide into	Write: $h = 32 - \text{prefix}$. Then 2^h and 2^{h-2}
Type 5 Protocol ID	name the protocols, which protocol is used, identify	Map scenario to Application layer, then Transport layer
Type 6 Explanation	explain, describe, write short note, differentiate	Structure: Definition → How it works → Example

The Mental Shift You Need**Stop reading to KNOW. Start reading to APPLY.**

After every concept you learn, immediately practice it on a real question.

Theory + Practice = Marks. Theory alone = 0 in BB AD Written.